

SQL Data Analysis: Hospital Management Case Study

10 Business Questions Solved Using PostgreSQL

1.Total Number of Patients

Business Question:

Write an SQL query to find the total number of patients across all hospitals.

SQL Query:

```
SELECT
    SUM (patients_count) AS total_patients
FROM hospital_data;
```

Business Insight:

This query calculates the total number of patients across all hospitals by summing the patient count for each hospital.

2.Average Number of Doctors per Hospital

Business Question:

Retrieve the average count of doctors available in each hospital.

SQL Query:

```
SELECT
    hospital_name,
    AVG (doctors_count) AS average_doctors
FROM hospital_data
GROUP BY hospital_name;
```

Business Insight:

This query calculates the average number of doctors available at each hospital, helping identify hospitals with higher or lower doctor availability and supporting resource planning and staffing decisions.

3.Top 3 Departments with the Highest Number of Patients

Business Question:

Find the top 3 hospital departments that have the highest number of patients.

SQL Query:

```
SELECT
    hospital_name,
    department,
    patients_count
FROM hospital_data
ORDER BY patients_count DESC
LIMIT 3;
```

Business Insight:

This query identifies the top three hospital departments with the highest patient counts, helping management understand where patient demand is highest and make informed decisions about staffing, capacity planning, and resource allocation.

4.Hospital with the Maximum Medical Expenses**Business Question:**

Identify the hospital that recorded the highest medical expenses.

SQL Query:

```
SELECT
    hospital_name,
    MAX (medical_expenses) AS
maximum_expenses
FROM hospital_data
GROUP BY hospital_name
ORDER BY maximum_expenses DESC
LIMIT 1;
```

Business Insight:

This query identifies the hospital with the highest medical expenses, helping management understand which hospital has the greatest financial expenditure and investigate the factors contributing to higher costs.

5.Daily Average Medical Expenses**Business Question:**

Calculate the average medical expenses per day for each hospital.

SQL Query:

```
SELECT
    hospital_name,
    AVG (
        medical_expenses /
        NULLIF (DATEDIFF (discharge_date, admission_date), 0)
    ) AS avg_expense_per_day
FROM hospital_data
GROUP BY hospital_name;
```

Business Insight:

This query calculates the average medical expense incurred per patient day at each hospital, helping management compare daily treatment costs and identify hospitals with relatively higher or lower healthcare expenses.

6.Longest Hospital Stay

Business Question:

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

SQL Query:

```
SELECT
    hospital_name,
    location,
    department,
    patients_count,
    admission_date,
    discharge_date,
    DATEDIFF (discharge_date, admission_date) AS stay_days
FROM hospital_data
ORDER BY stay_days DESC
LIMIT 1;
```

Business Insight:

This query identifies the patient record with the longest hospital stay, helping healthcare management understand extended hospitalization patterns and investigate factors that may contribute to longer patient stays.

7.Total Patients Treated Per City

Business Question:

Count the total number of patients treated in each city.

SQL Query:

```
SELECT
    location,
    SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY location
ORDER BY total_patients DESC;
```

Business Insight:

This query calculates the total number of patients in each location and ranks locations by patient volume, helping healthcare management identify areas with higher patient demand and plan resources, staffing, and hospital capacity accordingly.

8.Average Length of Stay Per Department**Business Question:**

Calculate the average number of days patients spend in each department.

SQL Query:

```
SELECT
    department,
    AVG (DATEDIFF (discharge_date, admission_date)) AS avg_stay_days
FROM hospital_data
GROUP BY department;
```

Business Insight:

This query calculates the average length of stay for patients in each department, helping hospital management identify departments with longer or shorter stays and improve capacity planning, staffing, and patient-care efficiency.

9.Identify the Department with the Lowest Number of Patients**Business Question:**

Find the department with the least number of patients.

SQL Query:

```
SELECT
    department,
    SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY department
```

```
ORDER BY total_patients ASC  
LIMIT 1;
```

Business Insight:

This query identifies the department with the lowest total patient volume, helping hospital management understand areas with lower demand and make informed decisions about resource allocation, staffing, and departmental capacity.

10.Monthly Medical Expenses Report

Business Question:

Group the data by month and calculate the total medical expenses for each month.

SQL Query:

```
SELECT  
    DATE_TRUNC('month', admission_date) AS month,  
    SUM(medical_expenses) AS total_medical_expenses  
FROM hospital_data  
GROUP BY month  
ORDER BY month;
```

Business Insight:

This query calculates the total medical expenses for each month, helping hospital management identify monthly spending trends, monitor changes in healthcare costs, and improve financial planning and budgeting.
